

Artificial Intelligence

First-Order Logic

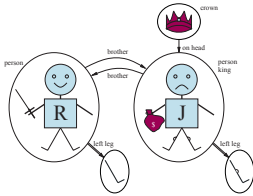
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1 Representation

Language	Ontological Commitment (What exists in the world)	Epistemological Commitment (What an agent believes about facts)
Propositional logic	facts	true/false/unknown
First-order logic	facts, objects, relations	true/false/unknown
Temporal logic	facts, objects, relations, times	true/false/unknown
Probability theory	facts	degree of belief $\in [0, 1]$
Fuzzy logic	facts with degree of truth $\in [0, 1]$	known interval value

2 Syntax and Semantics of First-Order Logic



3 Syntax and Semantics of First-Order Logic

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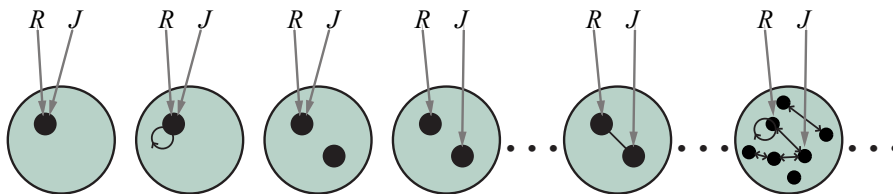
Sentence → AtomicSentence | ComplexSentence
AtomicSentence → Predicate( Predicate( Term, ... ) ) Term n Term
ComplexSentence → ( Sentence )
                | Sentence & Sentence
                | Sentence v Sentence
                | Sentence → Sentence
                | Quantifier Variable, ... Sentence

Term → Function( Term, ... )
      | Constant
      | Variable

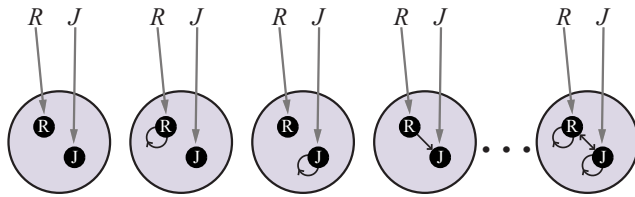
Quantifier → ∀ | ∃
Constant → a | b | ...
Variable → x | y | ...
Predicate → True | False | Atom | Relation | ...
Function → Mother | LeftLeg | ...

OPERATOR PRECEDENCE : ~, ∧, ∨, ∃, ∀, →, ⇔
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4 Syntax and Semantics of First-Order Logic



5 Syntax and Semantics of First-Order Logic



6 Using First-Order Logic

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7 Knowledge Engineering in First-Order Logic

